SENTINELAI

Real-Time AI Surveillance System

4 FPS	100%	0	<30ms
Detection Speed	Confidence Score	Training Required	WebRTC Latency

THE PROBLEM

Traditional surveillance is broken.

⚡ Slow Reaction

Human operators can't watch 24/7 — threats go unnoticed for minutes

❓ Rigid Detection

Classic CV models need thousands of labelled images to train

🔇 Silent Alerts

No real-time voice — operators must constantly check monitors

💸 Expensive Setup

Enterprise surveillance costs tens of thousands of dollars

THE SOLUTION

Introducing SentinelAI

An AI security guardian that watches your camera feed 24/7, detects threats instantly using plain English descriptions, and announces alerts via real-time voice — automatically.

Zero-Shot Detection

Describe any threat in English.
No training. No labelled data.

Voice Alerts

Gemini speaks threats aloud
the instant they're detected.

Real-Time Speed

Moondream + Stream WebRTC
for sub-30ms live detection.

AI Conversation

Ask SentinelAI questions.
Get instant voice responses.

HOW IT WORKS

Three-Layer Intelligence Architecture

01

Camera Feed Capture

Browser captures live video via Stream WebRTC

02

Moondream Detection

Zero-shot AI scans every frame for your custom threats

03

Gemini Voice Alert

Gemini Realtime announces threats aloud instantly

04

Dashboard Display

React UI shows bounding boxes & alert log

Powered by Stream's edge SFU network — ultra-low latency WebRTC connections worldwide

KEY FEATURES

Built for real-world security scenarios

Zero-Shot Detection

Detect ANY object using plain English — knife, fire, fallen person. No model retraining needed.

Instant Voice Alerts

Gemini Realtime shouts alerts the moment Moondream detects a threat with high confidence.

Bounding Boxes

Live green overlays on the SentinelAI video tile show exact threat location and confidence %.

Two-Way Voice

Talk to SentinelAI — ask questions, get status. Full conversational AI built in.

Auto-Reconnect

Session timeouts auto-recover. SentinelAI never goes offline during monitoring.

Custom Dashboard

Dark surveillance UI with dual video tiles, dynamic watchlist editor, and live alert log.

TECH STACK

Best-in-class tools, perfectly integrated

VISION AI

Moondream Cloud

Zero-shot object detection API.
Describe threats in plain English.

VOICE AI

Gemini 2.0 Realtime

Live voice + video understanding.
Two-way conversation built in.

VIDEO INFRASTRUCTURE

Stream WebRTC

Ultra-low latency SFU network.
333K minutes/month free tier.

AGENT FRAMEWORK

Vision Agents SDK

Open-source framework for
building real-time video AI agents.

BACKEND

FastAPI + Python

High-performance async backend.
Handles session management.

FRONTEND

React + Vite

Custom surveillance dashboard.
Dual video tiles + alert log.

LIVE DETECTION

SentinelAI in action

SENTINEL AI FEED

Live feed at 4 FPS

ALERT LOG

SYSTEM ONLINE 1:05:17 AM

SentinelAI surveillance active.

HIGH ALERT 1:06:32 AM

unattended bag detected! (1.00)

HIGH ALERT 1:08:14 AM

knife or blade detected! (0.87)

DETECTION 1:09:45 AM

person fallen on ground (0.72)

RESULTS

What we achieved in one hackathon session

100%

Confidence on unattended bag

4 FPS

Moondream detection speed

<30ms

Stream WebRTC latency

0

Hours of model training needed

- Zero-shot detection — knife, gun, fire, fallen person, unattended bag
- Two-way voice conversation with the AI security agent
- Auto-reconnect on session timeout — never goes offline
- Automatic Gemini voice alerts the moment threats are detected
- Beautiful React surveillance dashboard with dual video tiles
- 100% free tier APIs — accessible to every developer

SENTINELAI

Your AI Security Guardian

Built with ❤ using Stream Vision Agents SDK

github.com/kapilshastriwork-maker/sentinelai

Stream Vision Agents Hackathon 2026